Factor Detective: Puzzles with Factors and Multiples

Completing factor pairs and testing for primes, using divisibility rules, finding first common multiples, and extending number patterns

Your toolkit. Every puzzle below is solved by looking at how a number is built out of factors.

- **Factor pairs:** two numbers that multiply to make the target. A number with only the pair 1× itself is *prime*.
- **Divisibility rules:** a number is divisible by 2 if it is even, by 5 if it ends in 0 or 5, by 10 if it ends in 0, and by 3 if the *sum of its digits* is divisible by 3.
- **Multiples:** skip-count. The first number two skip-counts share is their first common multiple.

Show the factor pair, the digit sum, or the skip-count you used.

1. Complete the factor pair.

$$4 \times \_\_\_\_\_ = 24$$

Answer: ________________

2. Is 342 divisible by 3?
Add its digits, write the digit sum in the blank, and then answer yes or no.

$$3 + 4 + 2 = \_\_\_\_\_$$

Answer: ________________

3. Ari says 51 is prime because it is odd. Show that 51 is *composite* by writing a factor pair for it: $51 = 3\times$ what?

Answer: ________________

4. Pattern A starts at 4 and adds 4 each time. Pattern B starts at 6 and adds 6 each time. Write $<$ or $>$ to compare the 5th number of each pattern.

$$\text{5th number of A } \_\_\_\_ \text{ 5th number of B}$$

Answer: ________________

5. Skip-count by 6: $6, 12, \ldots$ Skip-count by 8: $8, 16, \ldots$
Keep going until a number appears in both lists. What is that first common multiple? Show which multiple of 6 and which multiple of 8 it is.

Answer: ________________

6. Is 63 a multiple of 5?
Fill in the skip-count that gets closest without passing 63, then finish the sentence.

$$5 \times 12 + 3 = \underline{\qquad}$$

Answer: ________________

7. Write 60 as a product of prime numbers only. Use a factor tree if it helps, and write the primes in order from smallest to largest.

Answer: ________________

8. A pattern begins $3, 7, 11, 15, \ldots$ and keeps adding the same amount.
What is the 8th number in the pattern? Explain how you found it without writing out every term.

Answer: ________________

9. **Riddle.** I am a number in the thirties. I am divisible by 3 and also by 4. Which number am I? Show why no other number in the thirties works.

Answer: ________________

10. Two bells ring together at the start of a lesson. One bell then rings every 8 minutes and the other rings every 12 minutes. How many minutes pass before they ring together again?

Answer: ________________ minutes

11. Ravi says 91 must be prime because it is odd and it is not in any times table he knows. (a) Find a factor pair for 91 and show that Ravi is wrong. (b) Explain why being odd does not make a number prime, and say how far you have to test for factors.

Answer: ________________

12. **Challenge.** A mystery number has exactly three factors: 1, itself, and one number in between. That in-between factor is 7.
(a) What is the mystery number? (b) Explain why a number with exactly three factors always has to be a prime multiplied by itself.

Answer: ________________